

Course Project: MNIST-1D

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Background

MNIST, treated in [LeCun et al. \(1998\)](#), is arguably the *Hello world* or *Drosophila melanogaster* (Fruit Fly) of Deep Learning. The MNIST (*Modified National Institute of Standards and Technology*) database comprises 70,000 images of handwritten digits, each 28x28 pixels of size. The task, most naturally, is to predict the handwritten digit as one of 10 classes.

MNIST has been, and still is, of great value for prototyping machine learning methods. However, it does display at least two important limitations. First, yet a very small dataset in the realm of modern computer vision, the 784 features of each image render it prohibitive for quick prototyping without access to a GPU or performant CPU. Second, it is, in some sense, *too easy* to solve. A (1-layer) linear classifier without preprocessing can achieve 88 % test accuracy ([LeCun et al., 1998](#)). Simple Multilayer-Perceptrons achieve roughly the same test performance as Convolutional Neural Networks (CNNs) (> 99 % test accuracy) ([Greydanus and Kobak, 2024](#)). The latter is irritating as CNNs, making use of structural information, are predestined to excel on image data.

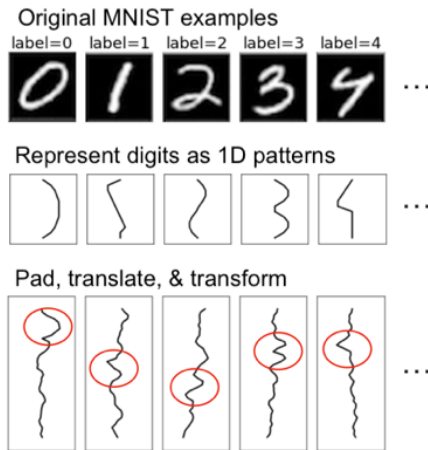
The MNIST-1D ([Greydanus and Kobak, 2024](#)) dataset was constructed to overcome these drawbacks. Its data is made of 40 features that represent 1 dimensional digit patterns. The dataset, by default consisting of 4,000 training and 1,000 test samples, is highly configurable and facilitates further data generation. This compact nature enables a multitude of experiments to be run in minutes on a standard laptop CPU. Furthermore, it succeeds in distinguishing model suitability to the task. MLPs score below 70 % while CNNs over 90 % in test accuracy.

Topic

It is your task to explore the MNIST-1D dataset in a way that is interesting, also to yourself, and present your findings in form of a poster. You are encouraged to come up with a research topic of your own but exemplary ones may be

- Evaluation of a specific model on the dataset. This model has to be different from the standard models already used in [Greydanus and Kobak \(2024\)](#) to which the model should be compared. The evaluation further has to contain comparisons to strong baselines.
- Tinkering with data preprocessing. The MNIST-1D repository makes augmentations to the dataset generation very accessible. You could test the effect of such preprocessing steps on the results of different models.

Constructing the dataset



Benchmarking common models

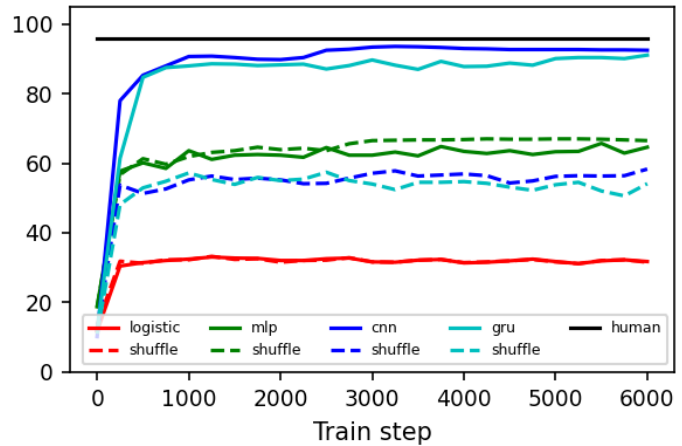


Figure 1: Constructing and benchmarking the MNIST-1D dataset. Taken from MNIST-1D's github repository: <https://github.com/greydanus/mnist1d>

- A study of dataset size on model performance.
- A study of feature size (deviating from the default 40 dimensions) on model performance.
- Comparison of results with those on actual image data. The original MNIST dataset may be too large for reasonable training but comparing with a subset of it, or sklearn's digits dataset (8x8 pixel images) should be feasible.

Other explorations can involve the phenomena that have already been studied in section 5 of Greydanus and Kobak (2024). In particular, you can reimplement and verify their findings on

- the Lottery ticket hypothesis and spatial inductive biases (section 5.1)
- deep double descent (section 5.2)
- gradient-based metalearning (section 5.3)
- metalearning an activation function (section 5.4)
- self-supervised learning (section 5.5)
- pooling methods (section 5.6)

For the latter 6 tasks, your exploration should **adapt, extend and compare** to the original implementation. That is, different (reasonable) settings and/or models should be used.

Poster

Poster requirements and evaluation have been adapted from [Martin Uray](#).

How to design a poster

Consult this short and very useful [guideline](#) for overall instructions. Videos on how to structure a good scientific poster are numerous, but we recommend [this one](#).

Various tools can be used to design a poster, but the most common are PowerPoint or other OS's presentation software and L^AT_EX. Here you find possible templates for [PowerPoint](#) and [L^AT_EX](#).

Poster Requirements

Your poster should be *at least* of size **DIN A3**. You can print large formats via the [university's printcenter](#) or at every copyshop.

Formal requirements

- Choose a suitable topic and title and include it in your poster
- Add an abstract to motivate your project (between 300-1000 characters)
- Add a paragraph with your key findings, support it with a table if feasible and interpret the results (between 500-1500 characters)
- include at least 2 different kinds of plots (created using Python) in your poster.
- Each plot should have a title, legend, labels, and should be self-explanatory
- Include your name and the course title on the poster
- Make sure the poster is visually appealing
- Optional: link or QR code to further information

Task

1. Communicate a proposal of your project idea at best before Christmas Break (December 18), but at the latest by **January 11, 2027, 18:00** via *blackboard*. Here, you lay out in 3-6 sentences what project topic you propose, why it is interesting, how you will study it and the manner in which you could visualize results. I will then either confirm your proposal or ask you to refine it.
2. Present your poster in a short video (3-6 minutes) that you uploaded together with the poster and the necessary code to produce the results on *blackboard* by **January 23, 2027, 23:59**. In the video, both you and the printed out poster should be visible. No additional material, e.g. slides, may be used.
3. Present your poster in-person to your fellow students and myself at our last session on **January 25, 2027, 12:00–15:00**. Here, I'll shortly ask questions about your poster, after which you should examine other student's posters.

Grading

The grading of the project will be based on several components and a combination of the following factors:

- Visual presentation using the poster
- Design and usage (variety) of plots, formal requirements of plots
- The topic of the project and the story told
- The short verbal presentation of the poster
- The ability to answer questions posed in the presentation

Important: Making use of an LLM, also via Copilot, is not permitted - neither for code needed to produce the results nor generation of the poster. If it is clearly indicated, that this rule has been violated, e.g. if a student is not able to explain how results or accompanying plots have been produced, the project will be failed with 0 points.

References

- Sam Greydanus and Dmitry Kobak. Scaling down deep learning with MNIST-1D. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://github.com/greydanus/mnist1d>.
- Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791.